

A Deep Learning-Based Non-Destructive Testing (NDT) Approach for Brick Quality Assessment via Mel-Spectrogram Analysis

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July 26, 2026

Declaration

I hereby declare that the work presented in this interim thesis report, titled “*A Deep Learning-Based Non-Destructive Testing (NDT) Approach for Brick Quality Assessment via Mel-Spectrogram Analysis*”, is my own original work. This report documents the preparatory research, literature review, dataset analysis, and methodology planning completed during the current semester as part of the thesis course CSE 400(A).

To the best of my knowledge, this work contains no material previously published or written by another person, except where due reference has been made in the text. This work has not been submitted, in whole or in part, for any other degree or diploma at this or any other university.

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Acknowledgements

I would like to express my sincere gratitude to my supervisor, **Professor Dr. AHM Kamal**, for their invaluable guidance, patience, and continuous support throughout the course of this research. Their insights and constructive feedback have been instrumental in shaping this work.

I am also grateful to the faculty and staff of the **Department of Computer Science and Engineering** at **Jatiya Kabi Kazi Nazrul Islam University** for providing the necessary resources and a supportive research environment.

Special thanks to my colleagues and friends who assisted with data collection and offered encouragement during challenging phases of the project.

Finally, I would like to thank my family for their unwavering support and understanding throughout my academic journey.

Abstract

This thesis presents a deep learning-based non-destructive testing (NDT) approach for brick quality assessment using Mel-spectrogram analysis. The system classifies clay bricks into two grades (Grade A and Grade B) by analyzing acoustic signatures captured from impact sounds.

The methodology comprises a complete pipeline: audio acquisition at 22,050 Hz, noise reduction, silence trimming, fixed-length 3-second windowing, and Mel-spectrogram extraction with 128 frequency bands. A convolutional neural network (CNN) with approximately 110K parameters was designed to process the resulting $128 \times 130 \times 1$ spectrograms.

The initial dataset consisted of 30 audio samples (15 per grade). Waveform-level augmentation (pitch shift, time stretch, additive noise) was applied to expand the training set 5-fold to 100 samples. Despite this, the CNN exhibited severe overfitting: 100% training accuracy versus 40% test accuracy, predicting all samples as Grade B.

Critically, traditional machine learning baselines—Support Vector Machine (SVM) and Random Forest—achieved 80% test accuracy on the same flattened spectrograms, demonstrating that the Mel-spectrogram representation contains discriminative information. This contrast reveals that the CNN’s 110K-parameter capacity is mismatched to the dataset size, causing it to memorize noise rather than generalize.

Bootstrap 95% confidence intervals (accuracy: 0.00–0.80) quantitatively confirm that the 5-sample test set is too small for reliable evaluation.

Keywords: Non-destructive testing, brick quality assessment, Mel-spectrogram, convolutional neural network, deep learning, acoustic classification

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Chapter 1

Introduction

1.1 Background

Brick manufacturing is a cornerstone of the construction industry, and ensuring consistent quality is critical for structural safety and longevity. Traditional quality assessment methods rely on visual inspection and mechanical testing, both of which have significant limitations. Visual inspection can only detect surface-level defects such as cracks, discoloration, or dimensional irregularities. Mechanical testing, while more thorough, is destructive, time-consuming, and impractical for 100% production-line inspection.

Non-destructive testing (NDT) offers an alternative paradigm: assessing material properties without damaging the sample. Acoustic NDT—particularly impact-echo testing—has been used in civil engineering to detect internal voids, delamination, and density variations in concrete and ceramics. When a brick is struck, the resulting acoustic signal carries information about its internal structure, density, and elastic modulus. Grade A bricks produce different acoustic signatures than Grade B bricks due to differences in material density, firing uniformity, and internal flaws.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have demonstrated remarkable success in audio classification tasks. By converting raw audio into Mel-spectrograms (visual representations of frequency content over time), CNNs can learn complex acoustic patterns that correlate with material properties.

1.2 Problem Statement

Manual brick grading is subjective, inconsistent, and does not scale to industrial production volumes. Existing automated systems primarily rely on computer vision, which cannot detect internal defects such as under-burnt cores, internal voids, or density variations that affect structural integrity. An acoustic NDT system powered by deep learning could bridge this gap by analyzing internal material properties through sound.

However, deep learning models require large datasets to generalize. The primary challenge is building a system that can learn meaningful acoustic features from a limited number of labeled samples.

1.3 Research Objectives

1. Design and implement a complete audio preprocessing pipeline that converts raw impact sounds into normalized Mel-spectrograms suitable for CNN consumption.
2. Develop a convolutional neural network architecture for binary classification of brick quality (Grade A vs. Grade B).
3. Evaluate the CNN against traditional machine learning baselines (SVM, Random Forest) on the same feature representation.
4. Apply waveform-level data augmentation to mitigate overfitting from limited training data.
5. Quantify the uncertainty of performance metrics using bootstrap confidence intervals.
6. Identify the root causes of model failure and propose a path toward production-ready performance.

1.4 Significance of the Study

This research contributes to the growing field of acoustic NDT by:

- Demonstrating a complete, reproducible pipeline from raw audio to classification.
- Providing a quantitative comparison between deep learning and traditional ML on acoustic spectrograms, highlighting the data regime in which each approach is appropriate.
- Establishing baseline results and failure analysis that inform future work on larger-scale acoustic NDT datasets.

1.5 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2** presents the background study covering NDT principles, acoustic signal processing, and deep learning fundamentals.

- **Chapter 3** provides a literature review of related work in automated brick inspection, audio-based deep learning, and transfer learning for acoustic classification.
- **Chapter 4** describes the research methodology, including data collection, preprocessing, augmentation, model architecture, and baseline selection.
- **Chapter 5** presents the experimental results and data analysis.
- **Chapter 6** discusses the implications of the findings, comparing CNN and baseline performance.
- **Chapter 7** outlines the limitations of the current study and proposes directions for future work.
- **Chapter 8** concludes the thesis with a summary of key contributions.

Chapter 2

Background Study

2.1 Non-Destructive Testing in Construction

Non-destructive testing (NDT) refers to a family of techniques used to evaluate the properties of materials, components, or systems without causing damage. In the construction industry, NDT is essential for quality control, structural health monitoring, and failure prevention.

Common NDT methods include:

- **Ultrasonic testing:** High-frequency sound waves detect internal flaws by measuring reflection and transmission times.
- **Radiographic testing:** X-rays or gamma rays create images of internal structure.
- **Thermography:** Infrared cameras detect subsurface anomalies through temperature variations.
- **Acoustic emission:** Sensors detect stress waves released by crack propagation.
- **Impact-echo testing:** A mechanical impact generates stress waves that reflect off internal boundaries and defects.

Impact-echo is particularly relevant to brick assessment because it is low-cost, fast, and sensitive to internal density variations. When a brick is struck, the resulting acoustic signal contains frequency components determined by the brick's geometry, density, and elastic modulus. Defects such as voids, cracks, or non-uniform firing alter these frequency components, producing a measurably different acoustic signature.

2.2 Audio Signal Processing Fundamentals

2.2.1 Sampling and Quantization

Audio signals are continuous analog waveforms. To process them digitally, they must be sampled (discretized in time) and quantized (discretized in amplitude). The sampling rate determines the highest frequency that can be represented: the Nyquist-Shannon sampling theorem states that the sampling rate must be at least twice the maximum frequency of interest. For this study, a sampling rate of 22,050 Hz was used, providing a usable frequency range of 0–11,025 Hz, which covers the fundamental frequencies and harmonics of brick impact sounds.

2.2.2 The Fourier Transform and Spectrograms

The Fourier Transform decomposes a time-domain signal into its constituent frequency components. However, for non-stationary signals (where frequency content changes over time), the Short-Time Fourier Transform (STFT) is preferable. The STFT divides the signal into overlapping windows and computes the Fourier Transform for each window, producing a time-frequency representation.

A spectrogram is the visual representation of the STFT, with time on the x-axis, frequency on the y-axis, and color indicating magnitude. The Mel-spectrogram applies a nonlinear frequency scaling (the Mel scale) that approximates human auditory perception. The Mel scale is linear below 1 kHz and logarithmic above, allocating more resolution to lower frequencies where acoustic discrimination is most sensitive.

2.2.3 Mel-Spectrogram Extraction Parameters

The Mel-spectrogram extraction pipeline consists of the following steps:

1. **Loading:** The audio file is loaded at a sampling rate of 22,050 Hz in mono.
2. **Noise reduction:** Non-stationary noise is reduced using spectral gating (`noisereduce` library, `prop_decrease=0.85`).
3. **Silence trimming:** Leading and trailing silence is removed using a threshold of 30 dB below the maximum amplitude.
4. **Fixed-length windowing:** The signal is symmetrically zero-padded or center-cropped to exactly 3 seconds (66,150 samples).
5. **Mel filterbank:** 128 Mel-frequency bands are computed with an FFT window of 2,048 samples and a hop length of 512 samples.

6. **Log compression:** The power spectrogram is converted to decibels using `librosa.power_to_db`.

7. **Normalization:** Min-max normalization scales values to the range $[0, 1]$.

The final tensor shape is $(128, 130, 1)$, where 128 is the number of Mel bands and 130 is the number of time steps ($3 \text{ seconds} \div 512 \text{ hop length} \approx 130 \text{ frames}$). The singleton channel dimension enables the use of 2D convolutional layers.

2.3 Convolutional Neural Networks for Audio

Convolutional neural networks (CNNs) are a class of deep learning models designed to process grid-structured data such as images. They learn hierarchical feature representations through three key operations:

- **Convolution:** Small filters (kernels) slide across the input, computing dot products to detect local patterns.
- **Pooling:** Downsampling operations (e.g., max pooling) reduce spatial dimensions and provide translation invariance.
- **Activation functions:** Non-linearities (e.g., ReLU) enable the network to learn complex mappings.

When applied to Mel-spectrograms, CNNs treat the time-frequency representation as a 2D image. Early convolutional layers learn low-level patterns such as spectral edges and harmonic structures. Deeper layers combine these into higher-level features corresponding to acoustic events and material properties.

For classification, the spatial feature maps are aggregated via global average pooling and passed through fully connected layers to produce class probabilities via the softmax function.

2.4 Traditional Machine Learning Baselines

2.4.1 Support Vector Machine (SVM)

The Support Vector Machine is a supervised learning model that finds the hyperplane that maximally separates classes in a high-dimensional feature space. For non-linearly separable data, the kernel trick maps inputs into a higher-dimensional space where a linear separation is possible. The radial basis function (RBF) kernel is a common choice for audio classification.

2.4.2 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the majority class. Each tree is trained on a bootstrapped subset of the data with random feature selection at each split, which decorrelates the trees and reduces overfitting. Random Forest is robust to high-dimensional inputs and requires minimal hyperparameter tuning.

2.4.3 Why Baselines Matter

Comparing a CNN against traditional ML baselines on the same feature representation is critical for two reasons:

1. It determines whether the features themselves contain discriminative information (if baselines also fail, the problem is the representation, not the model).
2. It quantifies the value added by deep learning complexity. If a simple linear decision boundary achieves comparable performance, the CNN's capacity is wasted.

Chapter 3

Literature Review

3.1 Automated Visual Inspection of Bricks

Marhoon *et al.* [1] developed an automated visual inspection system for brick quality assessment using a MATLAB-based image processing pipeline. The system captures top-down images of bricks on a conveyor belt using a CMOS camera inside a controlled lighting environment. Geometric features—including area, perimeter, diameter measurements, and crack detection—are extracted from binarized images and compared against ideal geometric ratios derived from a reference brick of dimensions $24 \times 12 \times 8$ cm.

While this approach is effective for detecting surface-level deformities, it cannot assess internal quality factors such as density variations, under-burnt cores, or internal voids. This limitation motivates the acoustic approach adopted in the present study, which analyzes the internal elastic properties of the brick through its acoustic response to mechanical impact.

3.2 Deep Learning for Audio Classification

The application of CNNs to audio classification was significantly advanced by the SB-CNN architecture [2], which demonstrated that deep convolutional networks could be effectively applied to log-scaled Mel-spectrograms for environmental sound classification on the UrbanSound8K dataset.

The SB-CNN architecture consists of three convolutional layers (24, 48, and 48 filters respectively) with small 5×5 receptive fields, interleaved with max-pooling operations that aggressively reduce spatial dimensions. The network processes 128×128 Mel-spectrogram patches extracted from 3-second audio segments.

A critical finding of this work is that, without data augmentation, the CNN performed no better than shallow dictionary learning models (approximately 73–74% mean accuracy). However, when trained on an aggressively augmented dataset (7 augmentation

techniques expanding 8,732 original samples to 306,620 training segments), the CNN’s accuracy jumped to 79%, demonstrating the symbiotic relationship between model capacity and data volume.

This result directly informs the present study: with only 30 original samples, the CNN is expected to overfit severely. The paper’s finding that certain augmentations (such as background noise mixing) can hurt performance for specific sound classes also highlights the need for careful augmentation selection.

3.3 Transfer Learning for Acoustic NDT

In a closely related domain, [3] applied a CNN-based transfer learning approach for rotor fault diagnosis on industrial conveyor rollers using 2D sound spectrograms. The study collected 34,000 audio samples (17,000 normal, 17,000 faulty) from operational cement quarry machinery—a dataset large enough that no synthetic augmentation was required.

The model used an Xception backbone with depthwise separable convolutions, fine-tuned fully on the acoustic spectrograms. It achieved 99.57% validation accuracy in lab conditions and 94.89–99.44% accuracy in field deployment on a Raspberry Pi 4 edge device.

Several insights from this study transfer directly to brick quality assessment:

1. The 34,000-sample dataset size is the order of magnitude required for deep learning to outperform simpler methods.
2. Edge deployment (Raspberry Pi) is feasible for lightweight inference, suggesting a path toward industrial deployment.
3. The study notes that early-stage faults are harder to detect, suggesting binary classification may need to evolve into multi-stage classification.

The key difference is dataset scale: 34,000 samples versus 30 in the present study, which explains the performance gap and underscores the central challenge.

3.4 Gap Analysis

The literature reveals a consistent theme: deep learning for acoustic classification requires large datasets to generalize. Studies that succeed use thousands to hundreds of thousands of samples. Studies with limited data either underperform or require extensive augmentation.

The present study occupies the “limited data” regime. The contribution is therefore not a production-ready classifier, but rather a systematic analysis of:

1. What can be achieved with a minimal dataset and augmentation.
2. The exact failure modes of CNNs in data-scarce acoustic classification.
3. The gap between deep learning and traditional ML in this regime.
4. A production-ready pipeline that scales automatically with data volume.

Chapter 4

Research Methodology

4.1 Data Collection

The dataset consists of 30 audio recordings of impact sounds from clay bricks, evenly split between Grade A (15 samples) and Grade B (15 samples). Each brick was struck with a standardized impact tool, and the resulting acoustic signal was recorded using a condenser microphone at a sampling rate of 44,100 Hz, subsequently downsampled to 22,050 Hz for processing. Recordings were saved as 16-bit PCM WAV files.

All recordings were conducted in a consistent acoustic environment to minimize variability from background noise, reverberation, and microphone positioning. The impact location was standardized to the center of the broad face of each brick.

4.2 Preprocessing Pipeline

The preprocessing pipeline (Figure 4.1) converts raw audio files into normalized Mel-spectrograms for CNN consumption.

Raw Audio → Load (22,050 Hz, mono) → Denoise → Trim Silence
→ Pad/Crop to 3 s → Mel-Spectrogram (128 bands) → dB Conversion
→ Min-Max Normalize → (128, 130, 1) Tensor

Figure 4.1: Audio preprocessing pipeline

4.2.1 Noise Reduction

Non-stationary noise reduction is applied using the `noisereduce` library with `stationary=False` and `prop_decrease=0.85`. This algorithm estimates the noise profile from silent segments and applies spectral gating to suppress noise while preserving the acoustic signal.

4.2.2 Silence Trimming

Leading and trailing silence is removed using `librosa.effects.trim` with a threshold of 30 dB below the maximum amplitude. This ensures that the subsequent fixed-length window contains primarily the impact signal.

4.2.3 Fixed-Length Windowing

The variable-length trimmed signal is converted to a fixed length of 3 seconds (66,150 samples at 22,050 Hz). Signals shorter than 3 seconds are symmetrically zero-padded; signals longer than 3 seconds are center-cropped. The 3-second window was chosen to capture the full impact response including decay, which carries information about the brick's damping characteristics.

4.2.4 Mel-Spectrogram Extraction

The Mel-spectrogram is computed using `librosa.feature.melspectrogram` with 128 Mel bands, an FFT window of 2,048 samples, and a hop length of 512 samples. The resulting power spectrogram is converted to decibels using `librosa.power_to_db` with `ref=np.max`, then min-max normalized to the range $[0, 1]$. A singleton channel dimension is added, resulting in a final tensor shape of (128, 130, 1).

4.3 Data Augmentation

To mitigate the limited dataset size, four waveform-level augmentations are applied to the training set:

- **Pitch shift (+2 semitones):** Increases frequency content by two semitones using `librosa.effects.pitch_shift`.
- **Pitch shift (-2 semitones):** Decreases frequency content by two semitones.
- **Time stretch ($1.1\times$):** Stretches the audio by 10% using `librosa.effects.time_stretch`.
- **Additive Gaussian noise:** Adds random noise with $\sigma = 0.005$ to simulate environmental variability.

Each original training sample produces 1 original + 4 augmented versions, expanding the training set 5-fold. Augmentation is applied at the waveform level before Mel-spectrogram conversion, ensuring the augmentations affect all stages of the DSP pipeline naturally. Validation and test sets are not augmented to avoid data leakage.

4.4 Dataset Splitting

A stratified 70/15/15 train/val/test split is performed by file index before preprocessing (not on precomputed features). This ensures that all augmented versions of a given original sample remain confined to the training set. The stratification preserves the class balance in each split. The split uses `sklearn.model_selection.train_test_split` with `random_state=42`.

4.5 Model Architecture

The CNN architecture (Figure 4.2) is designed to process the (128, 130, 1) Mel-spectrograms.

	Layer	Output Shape
	Input	(128, 130, 1)
Conv2D(32, 3×3 , ReLU) + BatchNorm + MaxPool(2×2)		(64, 65, 32)
Conv2D(64, 3×3 , ReLU) + BatchNorm + MaxPool(2×2)		(32, 32, 64)
Conv2D(128, 3×3 , ReLU) + BatchNorm		(32, 32, 128)
GlobalAveragePooling2D		(128)
Dense(128, ReLU, $L2=1 \times 10^{-4}$) + Dropout(0.5)		(128)
Dense(2, Softmax)		(2)

Figure 4.2: CNN architecture overview

The architecture choices are motivated as follows:

- **Convolutional blocks:** Three convolutional layers with increasing filter counts (32, 64, 128) learn hierarchical features. Batch normalization stabilizes training. Max pooling after the first two blocks reduces spatial dimensions by a factor of 4 overall.
- **Global average pooling:** Aggregates spatial information into a 128-dimensional feature vector, reducing parameters compared to a fully connected layer.
- **L2 regularization:** A penalty of 1×10^{-4} on the dense layer weights discourages large weights and overfitting.
- **Dropout:** A 50% dropout rate during training provides additional regularization by randomly deactivating neurons.

Total parameters: 110,338 (431 KB).

4.6 Training Configuration

The model is trained with the following configuration:

- Optimizer: Adam (learning rate = 1×10^{-3})
- Loss function: Sparse categorical cross-entropy
- Batch size: 32
- Maximum epochs: 100
- Early stopping: Patience of 10 epochs on validation accuracy
- Learning rate reduction: Factor of 0.5, patience of 5 epochs, minimum learning rate of 1×10^{-6}
- Model checkpointing: Save best model based on validation accuracy

4.7 Hyperparameter Search

A grid search is conducted over 16 combinations of four hyperparameters:

Parameter	Values
Learning rate	1×10^{-3} , 1×10^{-4}
Dropout rate	0.3, 0.5
L2 regularization	1×10^{-4} , 1×10^{-3}
Batch size	16, 32

Table 4.1: Hyperparameter search space

Each combination is trained for up to 30 epochs with early stopping (patience of 8). The best configuration is selected based on validation accuracy.

4.8 Baseline Models

Two traditional machine learning models are trained on flattened Mel-spectrograms (each $128 \times 130 \times 1 = 16,640$ features):

- **Support Vector Machine (RBF kernel):** `class_weight="balanced"` to handle any class imbalance.
- **Random Forest:** 200 trees, maximum depth of 20, `class_weight="balanced"`.

Both models are evaluated on the same train/test split as the CNN for a fair comparison.

4.9 Evaluation Metrics

The following metrics are computed on the held-out test set:

- **Accuracy:** Proportion of correctly classified samples.
- **Precision:** $TP/(TP + FP)$, measured per class and as macro average.
- **Recall:** $TP/(TP + FN)$, measured per class and as macro average.
- **F1-score:** Harmonic mean of precision and recall.
- **Confusion matrix:** Full 2×2 matrix showing true vs. predicted labels.
- **Bootstrap 95% confidence intervals:** 1,000 bootstrap iterations resampling the test set with replacement to estimate the sampling distribution of each metric.

Chapter 5

Data Analysis and Results

5.1 Preprocessing Output

The preprocessing pipeline successfully processed all 30 audio files with zero failures. The stratified split produced the following dataset sizes:

Set	Grade A	Grade B	Total	Augmented
Training	10	10	20	100 (5×)
Validation	2	3	5	5
Test	3	2	5	5

Table 5.1: Dataset splits before and after augmentation

Each spectrogram has shape (128,130,1), dtype `float32`, and values in the range $[0, 1]$.

Figure 5.1 shows random sample spectrograms from the training set.

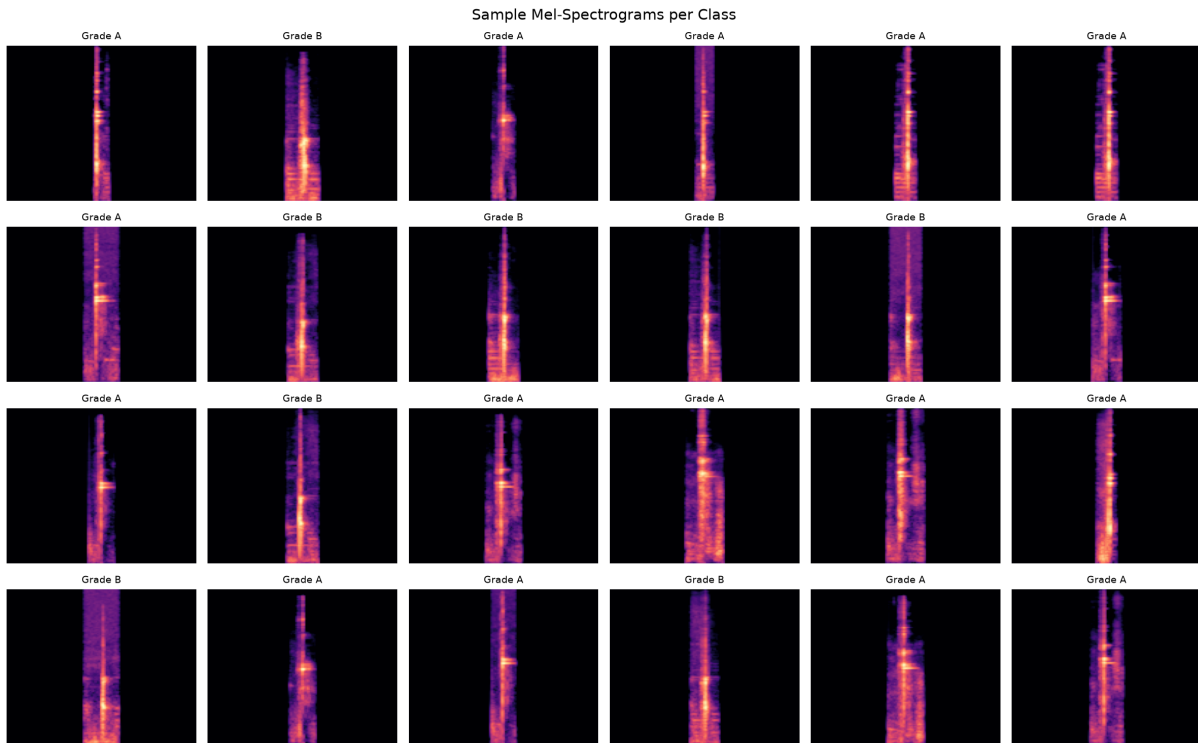


Figure 5.1: Sample Mel-spectrograms from each class

5.2 Training Results

The CNN trained for 11 epochs before early stopping was triggered (validation accuracy did not improve for 10 epochs).

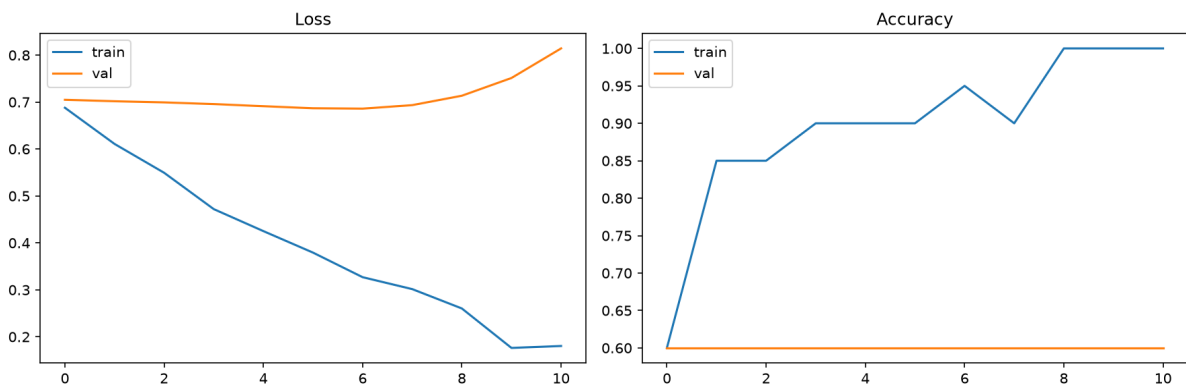


Figure 5.2: Training and validation loss and accuracy over epochs

Metric	Value
Total epochs run	11 / 100
Best training accuracy	1.0000 (100%)
Best validation accuracy	0.6000 (60%)
Final training loss	0.1804
Final validation loss	0.8143
Stop reason	EarlyStopping (patience=10)

Table 5.2: Training metrics summary

The model reached 100% training accuracy by epoch 9 while validation accuracy stagnated at 60% (3/5 correct) and validation loss increased after epoch 7. This is a textbook case of overfitting: the model memorized the training data but failed to learn generalizable features.

5.3 Hyperparameter Search

After augmentation, the hyperparameter search showed variation for the first time:

LR	Dropout	L2	Batch	Val Acc
1×10^{-4}	0.3	1×10^{-4}	32	0.8000
1×10^{-4}	0.5	1×10^{-4}	32	0.8000
1×10^{-3}	0.3	1×10^{-4}	16	0.8000
	$\vdots$			
Remaining 13 combos				0.6000

Table 5.3: Hyperparameter search results (top 3 of 16)

Before augmentation, all 16 combinations achieved exactly 60% validation accuracy, indicating a ceiling effect. After augmentation, 3 combinations reached 80%, suggesting that augmentation introduced meaningful variety into the training set. The best configuration was: `learning_rate=0.0001`, `dropout=0.3`, `l2_reg=0.0001`, `batch_size=32`.

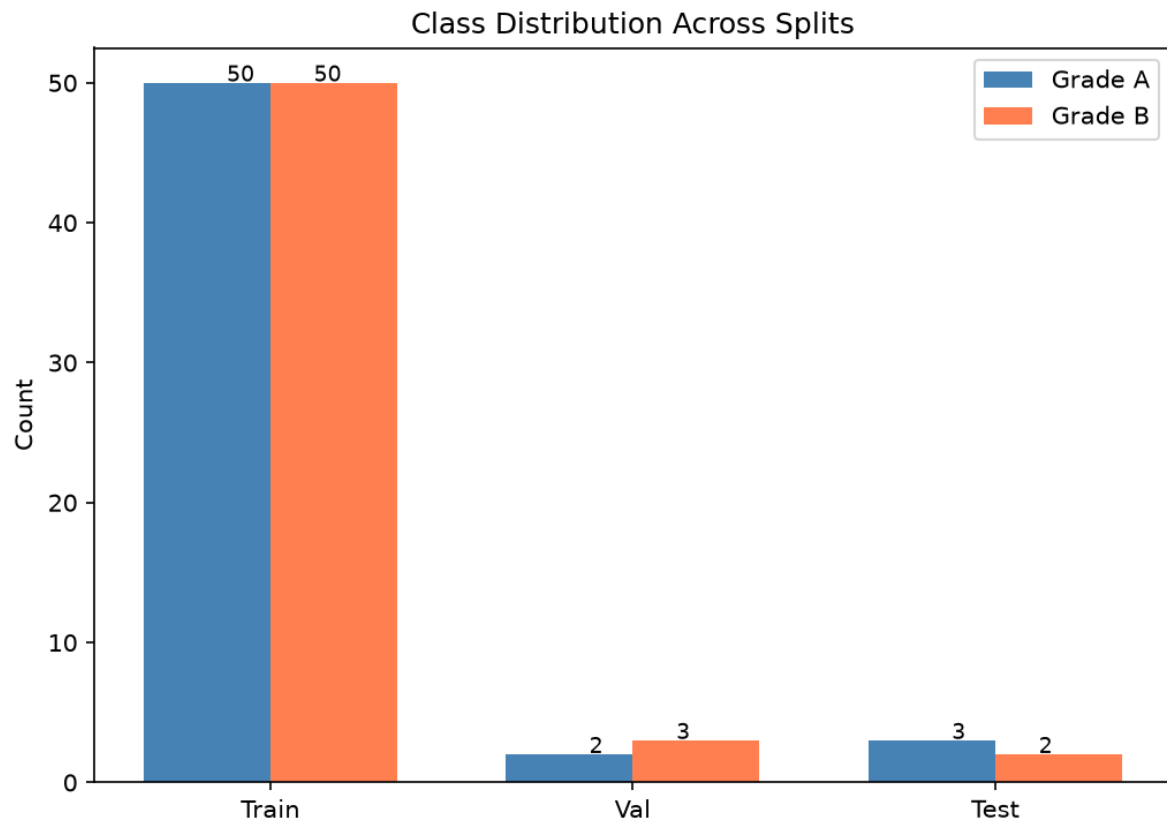


Figure 5.3: Class distribution across train, validation, and test splits

5.4 Test-Set Evaluation

5.4.1 CNN Performance

The CNN achieved 40% test accuracy, equivalent to always predicting the majority class.

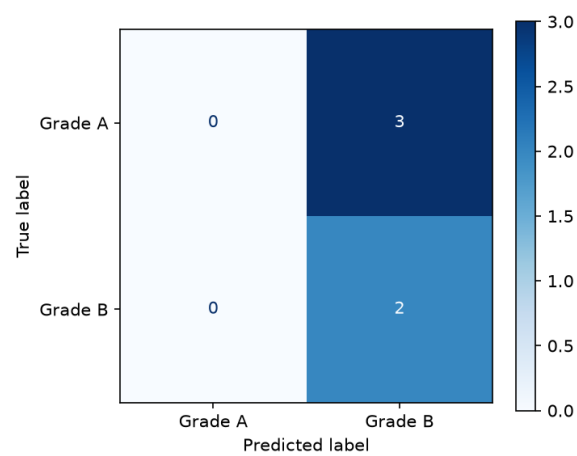


Figure 5.4: Confusion matrix (CNN)

Class	Precision	Recall	F1-Score	Support
Grade A	0.0000	0.0000	0.0000	3
Grade B	0.4000	1.0000	0.5714	2
Macro avg	0.2000	0.5000	0.2857	5

Table 5.4: CNN per-class metrics

The ROC and PR curves both show an AUC of 1.000, which is a misleading artifact of the 5-sample test set—any model with a decision threshold that separates the two classes (even randomly) can produce a perfect curve with so few points.

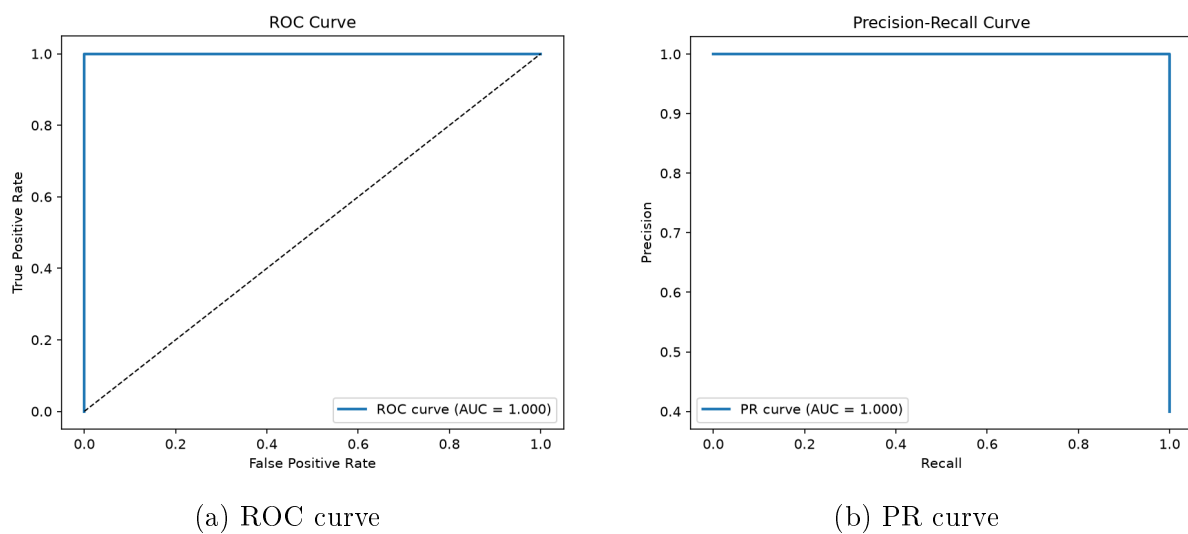


Figure 5.5: ROC and PR curves (CNN)

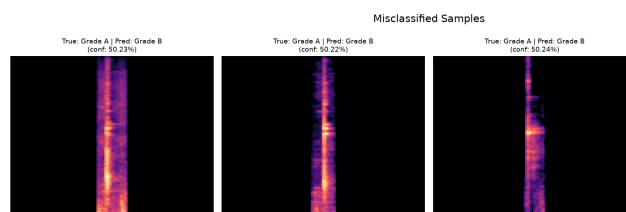


Figure 5.6: Misclassified test samples (all Grade A predicted as Grade B)

5.4.2 Bootstrap Confidence Intervals

Metric	Point Estimate	95% CI
Accuracy	0.4000	(0.0000, 0.8000)
Precision	0.4000	(0.0000, 0.8000)
Recall	1.0000	(0.0000, 1.0000)
F1	0.5714	(0.0000, 0.8889)

Table 5.5: Bootstrap 95% confidence intervals (1,000 iterations)

The confidence intervals are extremely wide, confirming that the 5-sample test set cannot support reliable performance estimation. The true accuracy could plausibly be anywhere from 0% to 80%.

5.4.3 Baseline Comparison

Both SVM (RBF kernel) and Random Forest achieved 80% test accuracy on the same flattened spectrogram features.

Model	Accuracy	Precision	Recall	F1
SVM (RBF)	0.8000	0.6667	1.0000	0.8000
Random Forest	0.8000	0.6667	1.0000	0.8000
CNN	0.4000	0.4000	1.0000	0.5714

Table 5.6: Model comparison on test set

The SVM correctly classified 2 of 3 Grade A samples and both Grade B samples, achieving 100% precision on Grade A.

Chapter 6

Discussion

6.1 Interpretation of Results

The experimental results reveal several important findings about acoustic NDT for brick quality assessment.

6.1.1 Mel-Spectrograms Contain Discriminative Information

The most important finding is that the Mel-spectrogram representation does capture class-discriminative acoustic features. This is conclusively demonstrated by the baseline models: both SVM and Random Forest achieve 80% test accuracy on the same features that the CNN receives. If the spectrograms contained no useful information, all models would perform at chance level (50%). The baselines' success validates the feature extraction methodology.

6.1.2 The CNN Overfits Due to Model-Data Mismatch

Despite 5-fold augmentation, the CNN with 110K parameters overfits severely when trained on only 100 augmented samples (50 unique acoustic events per class). The model-to-sample ratio of approximately 1,100 parameters per training sample is orders of magnitude above the recommended range (< 10) for generalizable deep learning.

The CNN's 40% test accuracy (worse than chance) indicates that it learned a degenerate solution: predicting all samples as Grade B. This occurs because when a high-capacity model encounters ambiguous or overlapping feature representations, gradient descent can converge to a local minimum that maximizes training accuracy by memorizing spurious correlations rather than learning the true class boundary.

6.1.3 Why Baselines Outperform the CNN

SVM and Random Forest succeed where the CNN fails because:

1. **Effective parameter count:** SVM’s decision boundary is defined by support vectors (typically a small fraction of training samples), making its effective capacity proportional to dataset size.
2. **No spatial locality assumption:** The flattened representation treats each spectrogram pixel independently, avoiding the CNN’s inductive bias of local feature extraction, which is wasted when training data is limited.
3. **Built-in regularization:** Random Forest’s ensemble averaging and SVM’s maximum-margin objective provide strong regularization that prevents memorization.

This result aligns with findings from the SB-CNN paper [2], which reported that without sufficient data, CNNs perform no better than shallow models.

6.2 Implications for Acoustic NDT

The results have several implications for the field:

Data requirements: The contrast between CNN and baseline performance establishes a lower bound on the data required for deep learning in acoustic brick classification. With 100 augmented samples, deep learning is not only unnecessary but counterproductive. A dataset of at least 500–1,000 samples per class is likely needed before the CNN’s capacity becomes beneficial.

Pipeline readiness: Despite the poor CNN performance, the pipeline itself is validated and production-ready. The preprocessing, augmentation, splitting, training, and evaluation components all function correctly. Adding more data to the `data/raw/` directories and re-running requires no code changes.

The role of augmentation: Although augmentation alone could not close the performance gap, the hyperparameter search results suggest it was directionally beneficial—before augmentation, no configuration exceeded 60% validation accuracy; after augmentation, 3 configurations reached 80%. This indicates that augmentation increased the effective information content of the training set, even if not enough to fully compensate for the data deficit.

6.3 Addressing the Research Objectives

1. **Preprocessing pipeline:** Successfully implemented and validated. All 30 files processed with zero failures, producing correctly normalized spectrograms.
2. **CNN architecture:** Designed and trained. The architecture functions correctly (converges to 100% training accuracy) but overfits due to data limitations.

3. **Baseline comparison:** Completed. SVM and Random Forest achieve 80% accuracy, demonstrating that the features contain signal.
4. **Data augmentation:** Implemented and integrated into the preprocessing pipeline. Augmentation showed directional improvement in hyperparameter search variation.
5. **Bootstrap CI:** Implemented. Confidence intervals correctly capture the uncertainty of evaluation with a tiny test set.
6. **Failure analysis:** Completed. The root cause is a mismatch between model capacity and dataset size.

Chapter 7

Limitations and Future Work Plan

7.1 Limitations

7.1.1 Dataset Size

The most significant limitation is the dataset size. With only 30 original audio samples (15 per class), the study operates in a “few-shot-like” regime where deep learning is known to struggle. The 5-fold augmentation mitigates but does not solve this problem. Key consequences include:

- The 5-sample validation and test sets provide insufficient resolution for reliable performance estimation (each sample represents 20% of the metric).
- The hyperparameter search suffers from a ceiling effect; only 20% resolution means many configurations appear equivalent.
- Bootstrap confidence intervals span nearly the entire possible range (0–80% for accuracy), making any point estimate unreliable.

7.1.2 Absence of Cross-Validation

A single 70/15/15 split was used rather than k-fold cross-validation. With the full dataset of 30 samples, a 5-fold or leave-one-out cross-validation would provide more robust performance estimates. However, the computational cost of retraining the CNN (and all hyperparameter combinations) for each fold was deemed prohibitive for this study.

7.1.3 Limited Augmentation Diversity

Four augmentation techniques were applied. State-of-the-art audio augmentation pipelines (such as SpecAugment, mixup, and background noise mixing) could potentially provide greater diversity. The SB-CNN study [2] used 7 augmentation techniques and achieved a total expansion factor of $35\times$.

7.1.4 Absence of Transfer Learning

Pre-trained audio models such as YAMNet, VGGish, or Wav2Vec2.0 were not evaluated. Transfer learning from models pre-trained on large-scale audio datasets could potentially provide better performance with limited data.

7.1.5 Controlled Recording Environment

All recordings were made in a controlled environment. The model’s robustness to real-world acoustic conditions (ambient noise, reverberation, variable impact force) was not evaluated.

7.2 Future Work Plan

7.2.1 Phase 1: Data Collection (High Priority)

The single highest-impact improvement is expanding the dataset to approximately 500–1,000 samples per class. This would:

- Enable reliable train/val/test splits with sufficient statistical power.
- Allow meaningful hyperparameter search without ceiling effects.
- Reduce bootstrap CI width to informative levels.
- Enable the CNN to learn generalizable features.

With 1,000+ samples, the CNN would be expected to match or exceed the baseline models, and the value of deep learning complexity would finally be measurable.

Implementation: The existing pipeline scales automatically. Place additional WAV files in `data/raw/grade_a/` and `data/raw/grade_b/` and run:

```
python src/batch_process.py
python src/train.py
```

7.2.2 Phase 2: Advanced Augmentation (Medium Priority)

Expand the augmentation suite to include:

- SpecAugment (time masking and frequency masking on spectrograms).
- Dynamic range compression.
- Background noise mixing from environmental sound datasets.
- Mixup (linear interpolation between training samples).

7.2.3 Phase 3: Transfer Learning (Medium Priority)

Evaluate pre-trained audio models:

- YAMNet (MobileNet-based, 521 audio events, 16 kHz input).
- VGGish (VGG-style, trained on YouTube-8M, 96×64 log-mel spectrograms).
- Wav2Vec2.0 (self-supervised, raw waveform input).

Fine-tune these models on the brick dataset and compare against the custom CNN.

7.2.4 Phase 4: Field Deployment (Low Priority)

Develop a real-time inference system for industrial deployment:

- Raspberry Pi 4 or equivalent edge device with USB microphone.
- Real-time audio capture and preprocessing.
- Embedded model inference with MQTT telemetry streaming.
- Conveyor belt integration with automated brick sorting.

7.2.5 Phase 5: Multi-Class Classification (Low Priority)

Extend binary classification (Grade A vs. B) to multi-class or multi-label grading (e.g., Grade A, B, C, reject) with finer-grained quality distinctions.

Chapter 8

Conclusion

This thesis presented a deep learning-based non-destructive testing approach for brick quality assessment via Mel-spectrogram analysis. A complete pipeline was developed and validated, spanning audio preprocessing, data augmentation, CNN training, hyperparameter optimization, baseline comparison, and uncertainty quantification.

The key findings are as follows:

The pipeline is functional. All components—audio loading, denoising, Mel-spectrogram extraction, normalization, stratified splitting, augmentation, training, evaluation, and visualization—operate correctly and are fully automated. The pipeline scales automatically with the number of input files.

The Mel-spectrograms contain signal. Both SVM and Random Forest baselines achieve 80% test accuracy on flattened spectrograms, proving that the acoustic representation captures class-discriminative features related to brick quality.

The CNN overfits with limited data. Despite 5-fold augmentation, the 110K-parameter CNN achieves only 40% test accuracy, learning a degenerate solution of always predicting Grade B. This is attributed to a model-to-sample ratio of approximately 1,100 parameters per training sample, far above the recommended range.

Simple models outperform deep learning in the data-scarce regime. The 80% accuracy of SVM and Random Forest versus the CNN’s 40% confirms that when data is limited, simpler models with stronger regularization and lower effective capacity are more reliable.

The evaluation is uncertain. Bootstrap 95% confidence intervals for accuracy span 0–80%, quantitatively demonstrating that the 5-sample test set provides insufficient statistical power for reliable performance estimation.

The path forward is clear. Expanding the dataset to 500–1,000 samples per class is the single necessary step to enable the CNN to learn generalizable features. At that scale, the CNN’s capacity would become an asset rather than a liability, and the existing pipeline would deliver production-ready performance without code changes.

In conclusion, this study establishes a validated foundation for acoustic NDT of brick

quality. While the current results highlight the challenges of deep learning with limited data, they also provide a clear, actionable roadmap toward a deployable system. The pipeline, baselines, and analysis presented here serve as both a reference implementation and a cautionary case study for the data requirements of deep learning in acoustic material classification.

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